

Antenna Datasheet

Product OC: YE0011BA
Version: 2.4
Date: 2026-05-27
Status: Released

Product Name: 4G Terminal Mount Rubber Monopole External Antenna
Key Features:
Frequency Band: 700–960 MHz, 1710–2170 MHz
Dimensions: Φ 10.5 mm × 50 mm × 16.7 mm
Efficiency: Up to 79.95 %
RoHS Compliant

Overview

YE0011BA is a 4G rubber antenna measuring Φ 10.5 mm × 50 mm × 16.7 mm. This ultra-wide-band 4G antenna provides broad coverage from 700–960 MHz, 1710–2170 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is terminated with SMA Male connector. This low profile, terminal mount omni-directional antenna, ideal for applications where the antenna is required to be discrete, is easy to install with maximum durability assured thanks to its TPE enclosure. It is compatible with Quectel's 4G Series modules.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. YE0011BA is designed as a monopole antenna, which needs to be mounted on a ground plane to offer high efficiency in all working bands. It is a perfect antenna product for customers that desire highest performance. This high-efficiency, high-gain omni-directional antenna is ideally suited for Zigbee, Bluetooth, IoT Sensors, public safety and smart safety, point of sales terminals, smart home automation, robotics / autonomous.

- **Typical applications include:**

✓ Zigbee
✓ Bluetooth
✓ IoT Sensors
✓ Public Safety and Smart Safety
✓ Point of Sales Terminals
✓ Smart Home Automation
✓ Robotics / Autonomous

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: On 130 mm × 130 mm EVB

1.1. Electrical

Electrical	
Frequency Range	700–960 MHz, 1710–2170 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional

Electrical – Detail

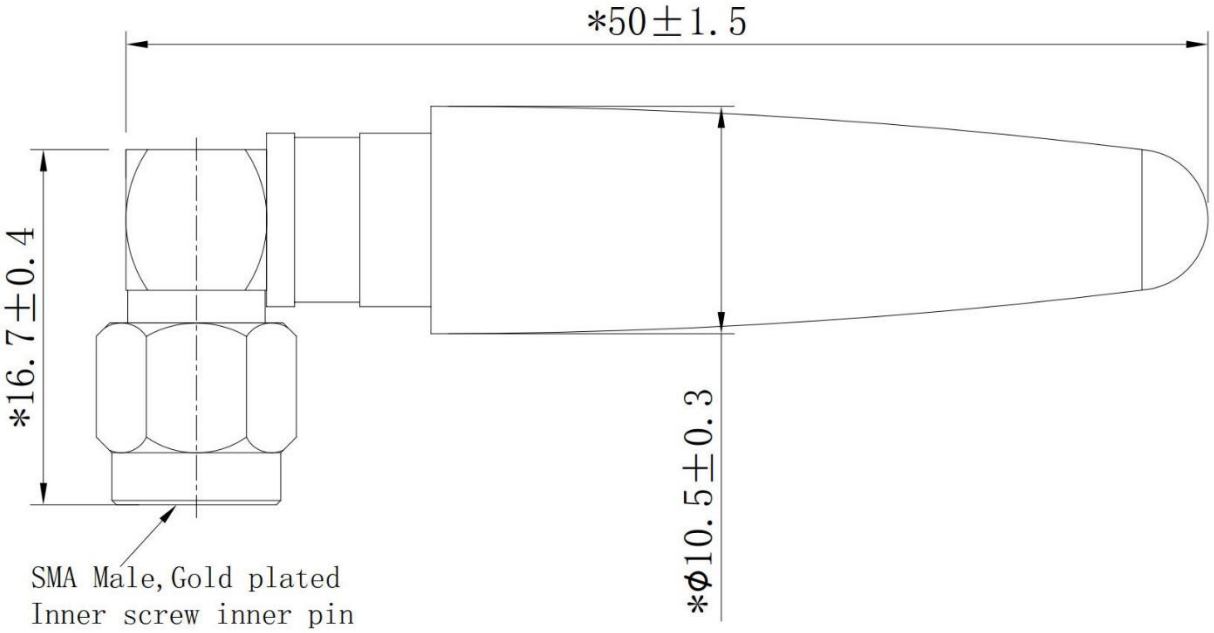
Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR		-	3.5	9.3	-	3.1	-	-	-	-	-	-
Max. Return Loss (dB)		-	-5.1	-1.9	-	-5.7	-	-	-	-	-	-
AVG Eff. (%)		-	71.6	50.9	-	61.3	-	-	-	-	-	-
AVG.AVG Gain (dB)		-	-1.5	-3.1	-	-2.1	-	-	-	-	-	-
Max. Peak Gain (dBi)		-	2.7	2.6	-	1.5	-	-	-	-	-	-

VSWR	≤ 9.3
Return Loss	≤ -1.9 dB
Peak Gain	≤ 2.7 dBi

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	Φ 10.5 mm × 50 mm × 16.7 mm
Casing Material & Color	TPE & Black
Connector Type	SMA Male
Mounting Type	Terminal
Weight	Typ. 6 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
RoHS Compliant	Yes

2 Drawing

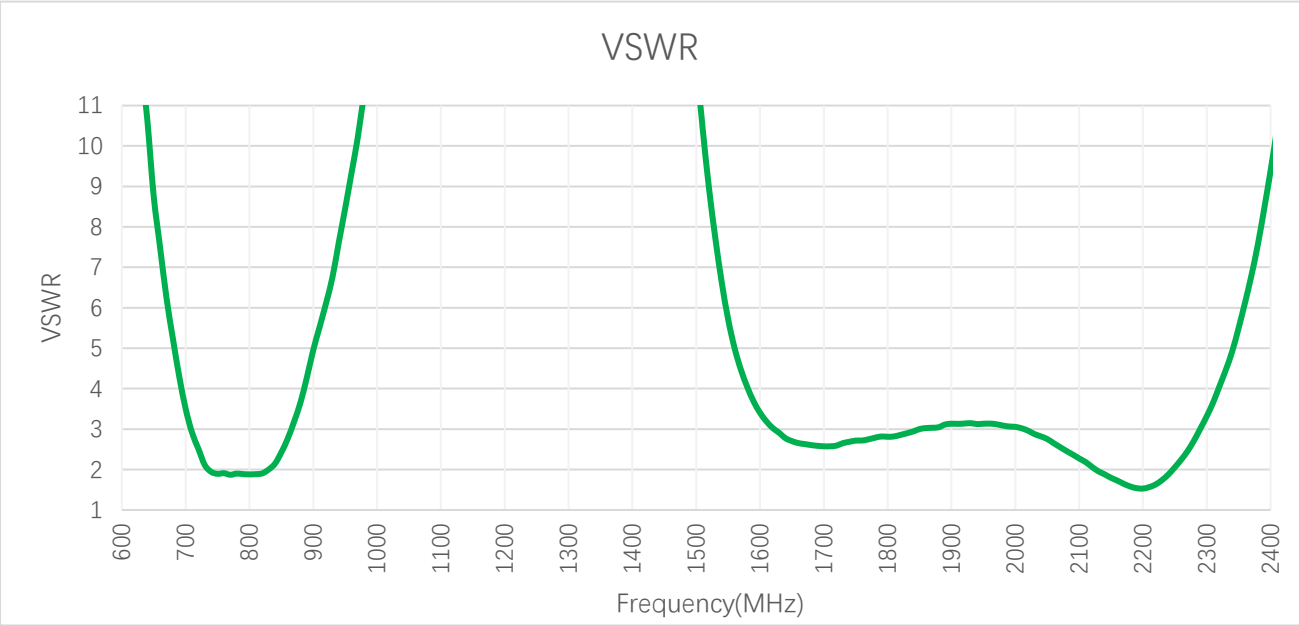

*50±1.5
*16.7±0.4
*φ10.5±0.3
SMA Male,Gold plated
Inner screw inner pin



3 Detailed Performance

3.1. S-Parameter Test

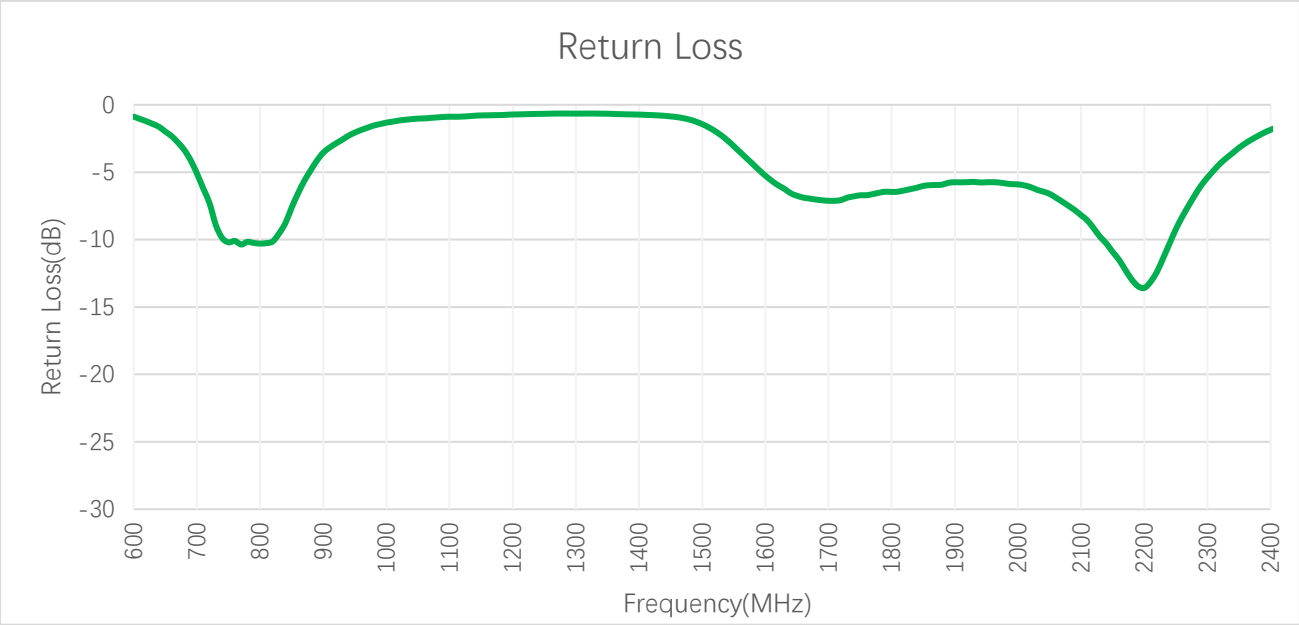
3.1.1. VSWR



VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
VSWR	-	-	2.9	2.0	5.0	9.3	-	2.6	2.7	3.0
Frequency (MHz)	1950	2140	2350	2450	2600	3600	4000	4700	5500	6000
VSWR	3.1	1.9	-	-	-	-	-	-	-	-

3.1.2. Return Loss

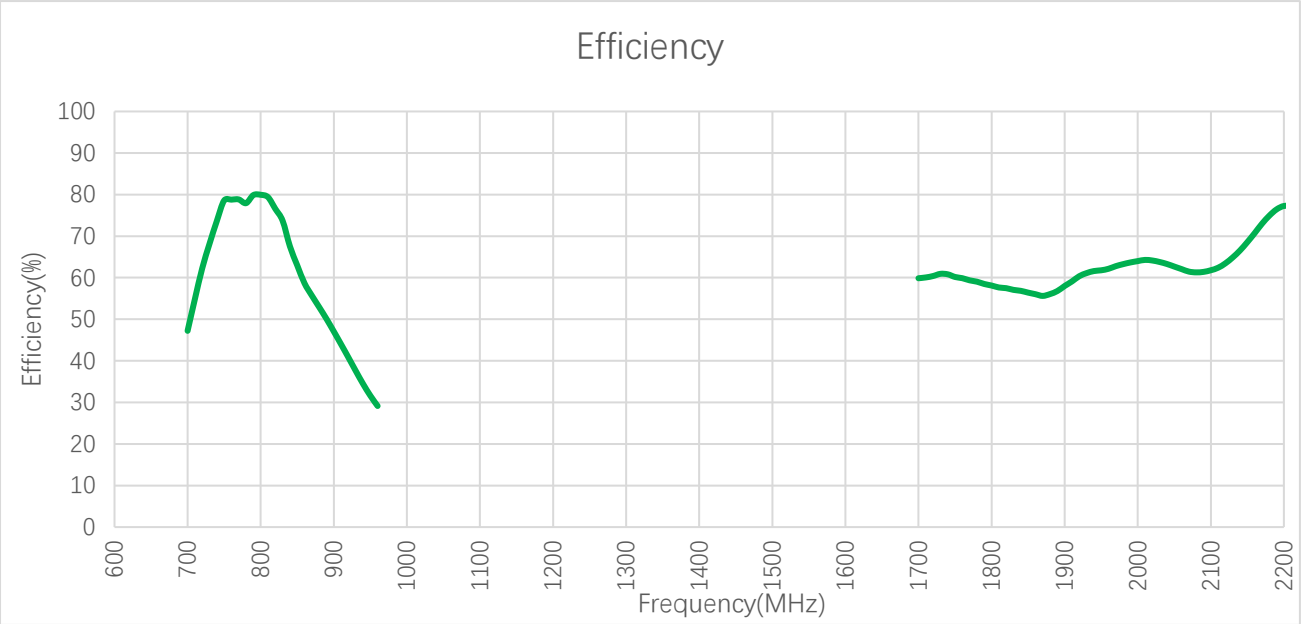


Return Loss (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Return Loss (dB)	-	-	-6.2	-9.5	-3.6	-1.9	-	-7.1	-6.8	-5.9
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4000**	**4700**	**5500**	**6000**
Return Loss (dB)	-5.7	-10.3	-	-	-	-	-	-	-	-

3.2. Radiation Performance Test

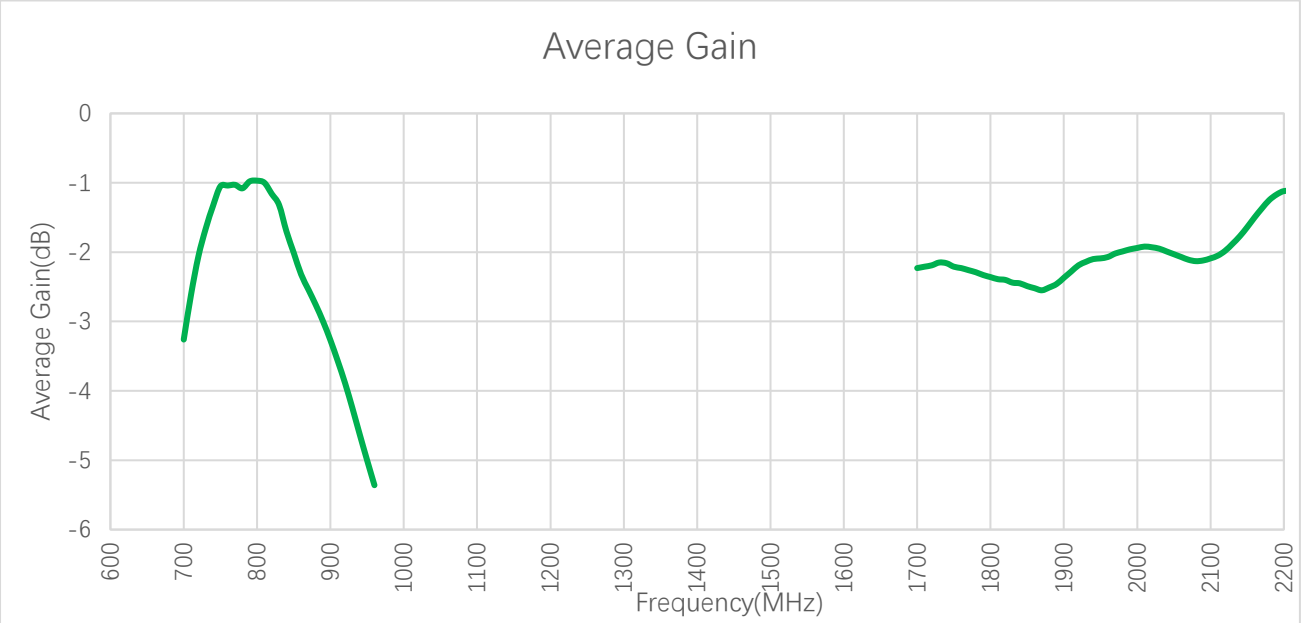
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	**600**	**630**	**710**	**830**	**900**	**960**	**1440**	**1710**	**1740**	**1880**
Efficiency (%)	-	-	54.9	73.7	47.1	29.1	-	60.1	60.8	56.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4000**	**4700**	**5500**	**6000**
Efficiency (%)	61.8	66.6	-	-	-	-	-	-	-	-

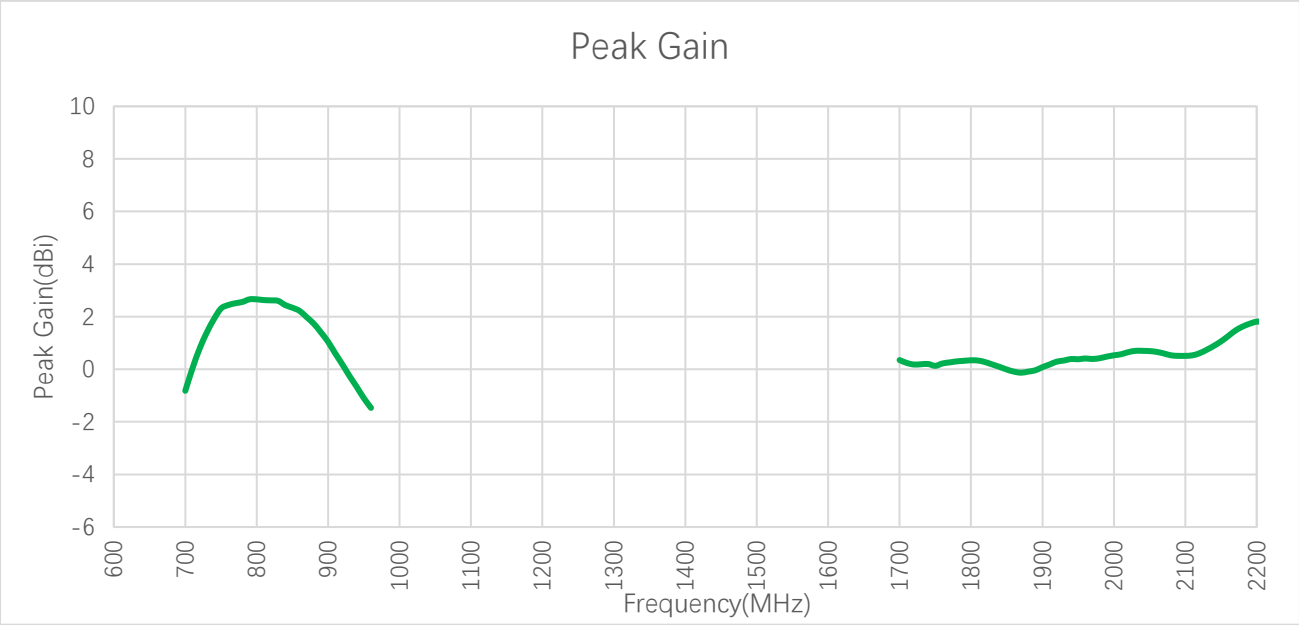
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Average Gain (dB)	-	-	-2.6	-1.3	-3.3	-5.4	-	-2.2	-2.2	-2.5
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4000**	**4700**	**5500**	**6000**
Average Gain (dB)	-2.1	-1.8	-	-	-	-	-	-	-	-

3.2.3. Peak Gain



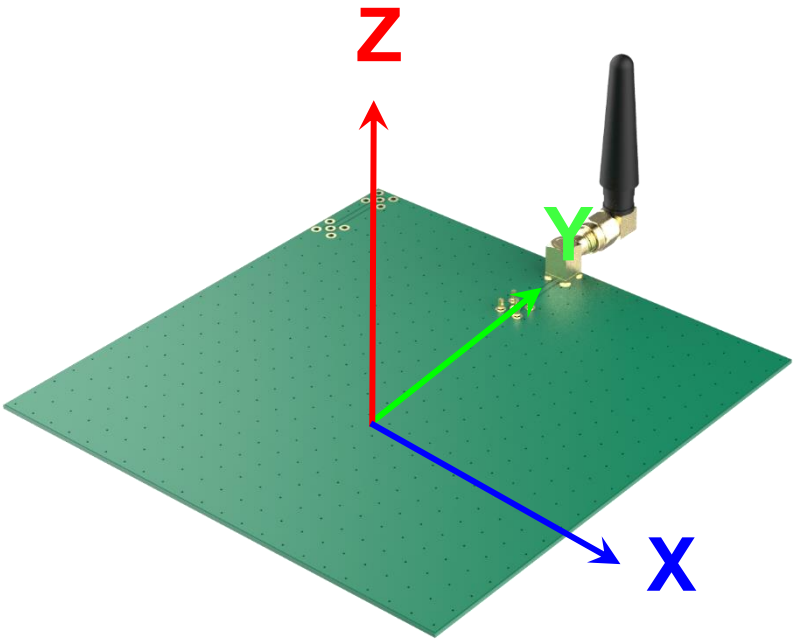
Peak Gain (dBi)

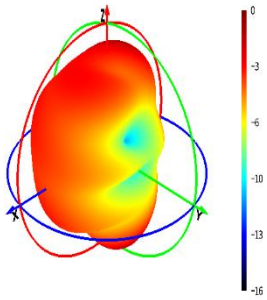
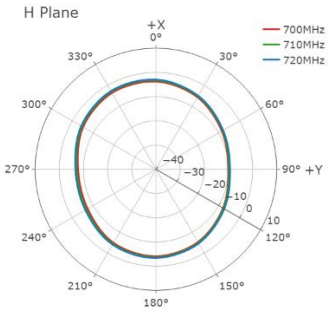
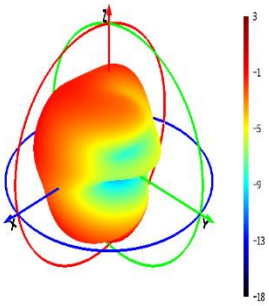
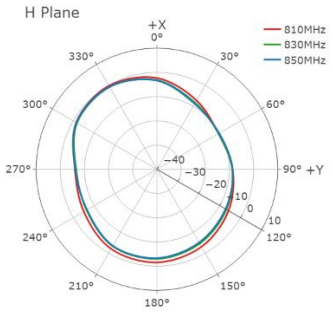
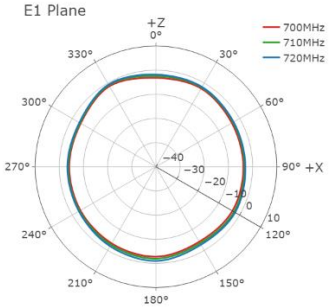
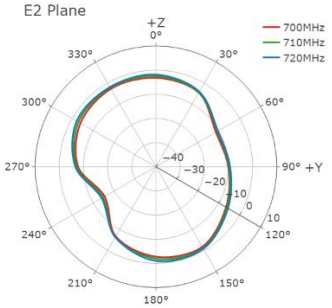
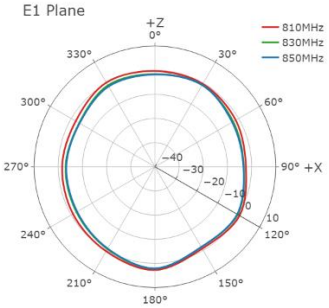
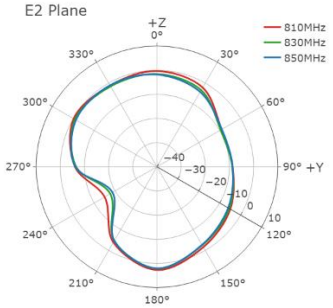
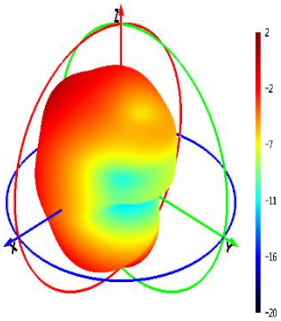
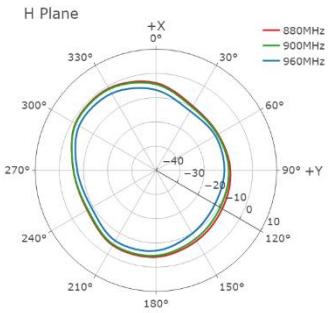
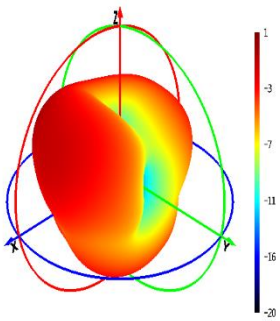
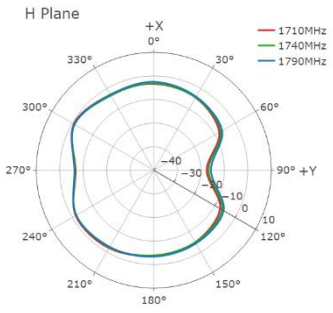
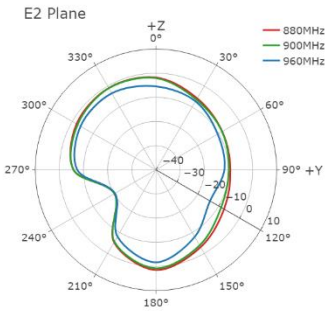
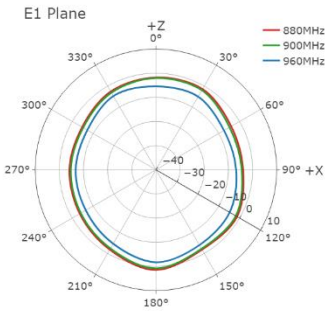
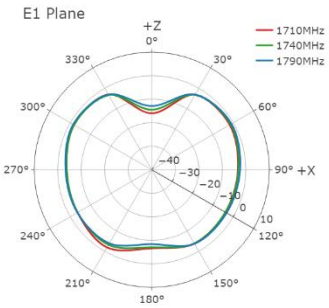
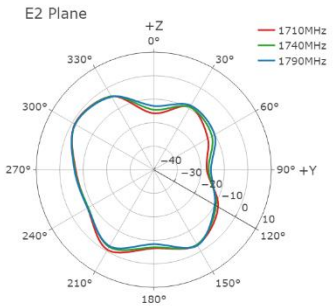
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Peak Gain (dBi)	-	-	0.0	2.6	1.0	-1.5	-	0.2	0.2	-0.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4000**	**4700**	**5500**	**6000**
Peak Gain (dBi)	0.4	0.9	-	-	-	-	-	-	-	-

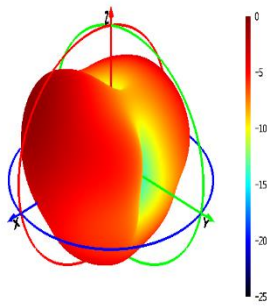
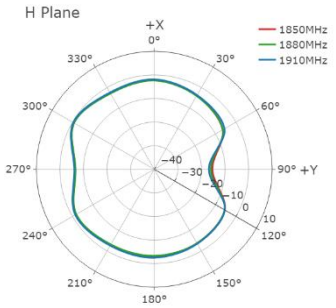
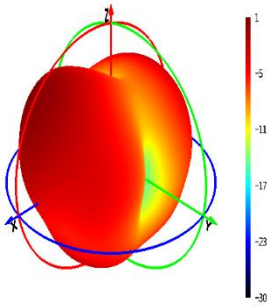
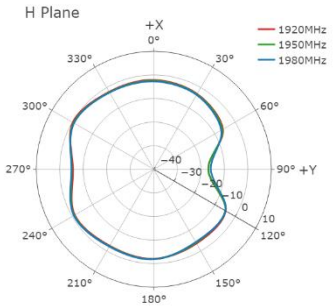
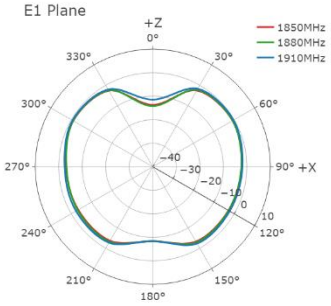
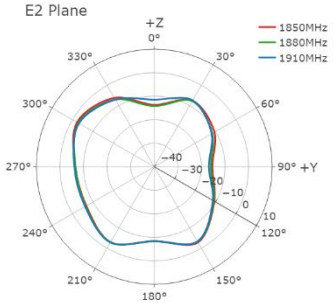
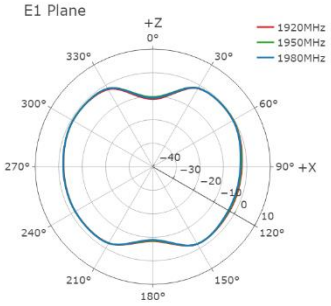
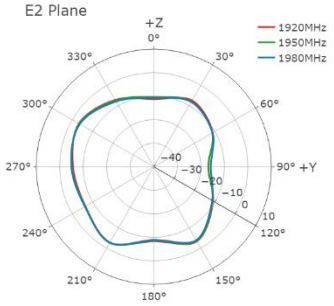
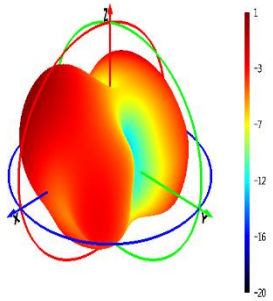
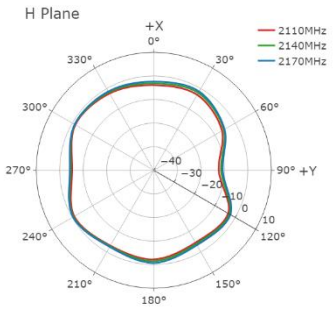
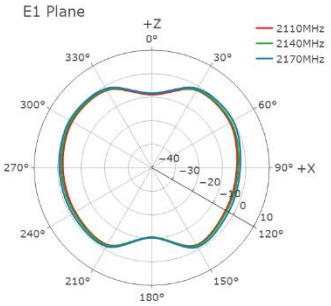
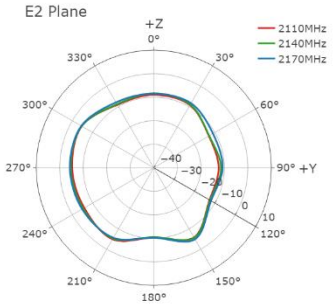
3.2.4. 3D & 2D Radiation Pattern

3.2.4.1. Test Condition: On 130 mm × 130 mm EVB

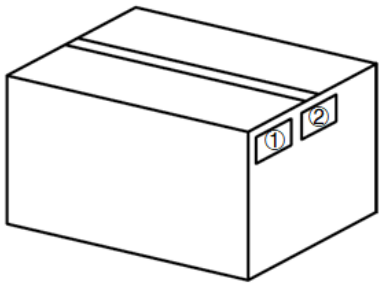
- Test Chamber: GL-S-1

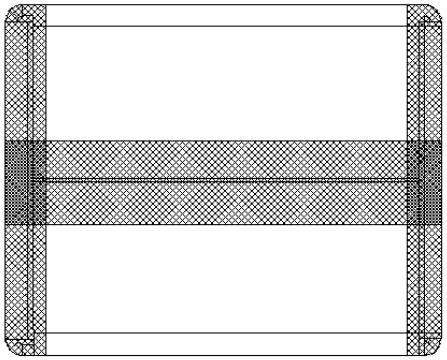


710 MHz		830 MHz	
			
			
900 MHz		1740 MHz	
			
			

1880 MHz		1950 MHz	
			
			
2140 MHz			
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		10 Antenna products in a One-piece bag. (10 Antennas / One-piece Bag)
2		Put 50 products into a PE bag.
3		(30 PE Bags / Carton Box) (1500 Antennas / Carton Box) Estimated quantity Products that are not full will be packaged in suitable cardboard boxes. Carton Size: L × W × H = 405 × 293 × 185 mm
4		**Position for Attaching Labels** ① Carton Label ② Quality Label

5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
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Or email us at: support@quectel.com.

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Revision History

Version	Date	Author	Note
-	2022-06-08	Ezail Tan	Creation of the document
1.0	2022-06-29	Ezail Tan	First official release
1.1	2022-09-09	Ezail Tan	Update test data.
1.2	2022-10-17	Aria Chu	Updated the IP rating.
2.0	2023-04-27	Joyful Huang/ Lucky Feng/ David Liu/ Aria Chu	Updated all data and datasheet template.
2.1	2025-02-08	Aria Chu	Deleted storage information and added storage temperature (Chapter 1.2).
2.2	2025-04-09	Allow Xu	1. Updated the overview. 2. Updated the packaging (Chapter 4).
2.3	2025-06-12	Lance Sun	1. Removed the IP53 waterproof rating (Chapter 1.2). 2. Updated the drawing (Chapter 2).
2.4	2026-05-27	Toby Wang/ Kessie Geng	1. Updated the casing material (Chapter 1.2). 2. Updated the drawing (Chapter 2). 3. Updated the packaging (Chapter 4).



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